

PHYSICS • UPPER SIXTH

UPPER SIXTH — PHYSICS

CHAPTER 6

ELECTROMAGNETIC WAVES

SUGGESTED TIME: 9 PERIODS

IN THIS CHAPTER

This chapter introduces electromagnetic waves, their spectrum, production and uses, then develops the wave behaviour of light: diffraction and interference with gratings, reflection, refraction, dispersion, lenses, prisms and optical instruments. It closes with the energy of waves in free space and the inverse square law, preparing students for GCE Advanced Level Paper 2 and Paper 3.

- By the end of this chapter I can describe an experiment to measure the speed of sound in air and compare it with the speed of light.
- By the end of this chapter I can state the characteristics of electromagnetic waves and order the regions of the EM spectrum by wavelength and frequency.
- By the end of this chapter I can explain the production, detection and uses of each region of the EM spectrum, including X-rays.
- By the end of this chapter I can explain plane polarisation of EM waves and give its applications.
- By the end of this chapter I can use the grating equation $d \sin \theta = n\lambda$ to determine wavelengths and analyse multiple-slit diffraction.
- By the end of this chapter I can state and apply the laws of reflection and refraction and use Snell's law with refractive index.



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CHAPTER OBJECTIVES

- By the end of this chapter I can describe an experiment to measure the speed of sound in air and compare it with the speed of light.
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- By the end of this chapter I can state and apply the laws of reflection and refraction and use Snell's law with refractive index.
- By the end of this chapter I can explain dispersion, total internal reflection and the critical angle, with applications such as optical fibres.
- By the end of this chapter I can use the thin lens formula and magnification to predict images formed by a converging lens, and use the diopetre.
- By the end of this chapter I can describe the passage of light through a prism and the working of a compound microscope and an astronomical telescope.

- By the end of this chapter I can apply conservation of energy to a point source and use the inverse square law for intensity.

PREREQUISITES

- Wave basics from Lower Sixth: amplitude, wavelength, frequency, period, $v = f\lambda$
- Trigonometry: sine, tangent, small-angle approximations
- Reflection of light at plane mirrors and image formation (Form 5 / Lower Sixth)
- Simple harmonic motion and progressive waves (Lower Sixth)
- Basic electricity: potential difference, electron acceleration (useful for X-ray production)

1 SECTION 1: MEASURING THE SPEED OF SOUND AND THE NATURE OF ELECTROMAGNETIC WAVES

Every evening in Yaoundé, CRTV broadcasts reach remote villages, hospital X-ray machines in Douala reveal broken bones without surgery, and a fisherman on the Wouri estuary sees a fish that is not exactly where it appears. All these phenomena — radio signals, X-rays, the bending of light in water, the rainbow after a tropical downpour — are governed by one family of waves: **electromagnetic waves**. How can the same physical entity carry a phone call, cook food in a microwave, and let a surgeon see inside the human body? And why does a straw in a glass of palm wine look broken at the surface? This chapter answers these questions by exploring the nature, propagation, and manipulation of electromagnetic waves, with light as our main laboratory.

Before we can handle electromagnetic waves with confidence, we must sharpen our tools on a wave we already know well: **sound**. Measuring the speed of sound in air is a classic GCE practical, and it teaches us the universal wave relation $v = f\lambda$ in a concrete setting. We then contrast sound with light and arrive at Maxwell's beautiful picture of the electromagnetic wave.

1.1 Recap of Progressive Waves

DEFINITION Progressive wave

A **progressive wave** is a disturbance that travels through a medium (or through vacuum, for electromagnetic waves), transferring energy from one point to another without any net transfer of matter.

Think of the “wave” that travels round a football stadium in Douala: each spectator simply stands up and sits down, yet the disturbance travels right round the stand. The spectators are the medium; only energy and information move.

Three quantities describe any progressive wave:

- the **wavelength** λ : the shortest distance between two points vibrating in phase (e.g. two successive crests), measured in metres;
- the **frequency** f : the number of complete vibrations per second made by any point of the medium, measured in hertz (Hz);
- the **speed** v : the distance travelled by the disturbance per unit time, in m s^{-1} .

PROPERTY The wave equation

For any progressive wave,

$$v = f\lambda.$$

Justification. In one period $T = 1/f$, the source completes exactly one oscillation and the dis-

turbance advances by exactly one wavelength λ . Hence

$$v = \frac{\text{distance}}{\text{time}} = \frac{\lambda}{T} = f\lambda.$$

Sound in air is a **longitudinal** wave: the air molecules oscillate parallel to the direction of propagation, producing alternating compressions and rarefactions. Light, as we shall see, is **transverse**. The relation $v = f\lambda$ applies to both.

1.2 Measuring the Speed of Sound in Air

Method 1: The echo method.

EXPERIMENT The echo (clap–echo) experiment

Stand at a measured distance d (e.g. 100 m) from a large flat wall on the school field. Clap hands (or strike two wooden blocks) repeatedly and adjust the rhythm until each clap coincides exactly with the echo of the previous one. A partner times $n = 50$ clap intervals with a stopwatch, giving the total time t . The time for one round trip is t/n , and the sound travels $2d$ in that time:

$$v = \frac{2d}{t/n} = \frac{2nd}{t}.$$

Sources of error: reaction time of the timekeeper (reduced by timing many claps), wind, and uncertainty in judging coincidence of clap and echo.

Method 2: The resonance tube.

This is the precise laboratory method. A long glass tube, open at the top, is connected at its base to a reservoir so that the water level — and hence the length l of the air column — can be varied. A vibrating tuning fork of known frequency f is held above the open end. The fork sends sound down the tube; the wave reflects at the water surface (a closed end) and a **stationary wave** forms in the air column. When the column length is just right, the air vibrates strongly and a loud sound is heard: this is **resonance**.

At resonance the air column has a node at the water surface and an antinode just above the open end. The distance from the open end to the antinode is the **end correction** e . Starting from a high water level and lowering it slowly, the first (shortest) resonance occurs at length l_1 and the second at length l_2 :

$$l_1 + e = \frac{\lambda}{4}, \quad l_2 + e = \frac{3\lambda}{4}.$$

Subtracting eliminates the unknown end correction:

$$l_2 - l_1 = \frac{\lambda}{2} \quad \Rightarrow \quad \lambda = 2(l_2 - l_1).$$

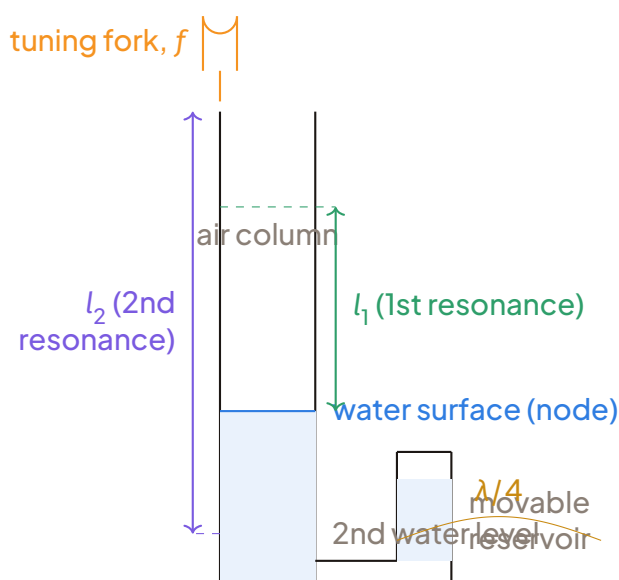
METHOD Speed of sound by the resonance tube

1. Strike the tuning fork of known frequency f and hold it over the open end of the tube.

- Lower the water level slowly from the top; note the length l_1 of the air column at the first (loudest) resonance.
- Continue lowering the water; note the length l_2 at the second resonance.
- Compute the wavelength: $\lambda = 2(l_2 - l_1)$. The end correction cancels automatically.
- Compute the speed:

$$v = 2f(l_2 - l_1)$$

- Repeat with forks of different frequencies and average, or plot v values, to improve accuracy. Note the air temperature, as v varies with $\sqrt{T}$.



Resonance tube: the water level is adjusted until the air column resonates with the tuning fork. The first and second resonance lengths satisfy $l_2 - l_1 = \lambda/2$. The standing wave shown corresponds to the first resonance (quarter-wave).

WORKED EXAMPLE 1 Resonance tube calculation

A tuning fork of frequency $f = 512$ Hz is held over a resonance tube. Resonances are heard when the air column is $l_1 = 16.0$ cm and $l_2 = 49.2$ cm long. Calculate the speed of sound in air.

SOLUTION

Solution. Convert lengths to SI units: $l_1 = 0.160$ m, $l_2 = 0.492$ m.

$$\lambda = 2(l_2 - l_1) = 2(0.492 - 0.160) = 2 \times 0.332 = 0.664 \text{ m.}$$

$$v = f\lambda = 512 \times 0.664 \approx 340 \text{ m s}^{-1}.$$

This agrees well with the accepted value of about 340 m s^{-1} at room temperature.

WARNING Common mistakes with the resonance tube

- Do not write $\lambda = 4l_1$ and stop there: the antinode lies slightly above the open end, so $l_1 + e = \lambda/4$ contains the unknown end correction e . Using l_1 alone gives a systematically low value of v .
- The clean formula is $v = 2f(l_2 - l_1)$: subtracting the two resonance positions eliminates the end correction.
- Do not confuse the first resonance ($\lambda/4$) with the second ($3\lambda/4$); the difference is $\lambda/2$, not λ .
- Quote v in m s^{-1} : convert centimetres to metres before substituting.

METHOD Exam tip (GCE): describing the echo experiment

If asked to describe an experiment to measure the speed of sound by the echo method, structure your answer in four parts:

1. Labelled diagram: observer, measuring tape, distance d , large reflecting wall.
2. Measurements: distance d with a tape; time t for n clap–echo intervals with a stop-watch, clapping in rhythm so each clap coincides with the previous echo.
3. Calculation: $v = \frac{2nd}{t}$ (the sound makes a return journey, hence the factor 2).
4. Sources of error and precautions: reaction time (time many intervals), wind along the path, choice of a large flat wall, repeat and average.

1.3 Sound versus Light: Two Kinds of Wave

Why do we bother contrasting sound and light? Because the contrast reveals what is truly remarkable about electromagnetic waves.

Feature	Sound	Light (EM wave)
Nature	Mechanical wave	Electromagnetic wave
Medium	Needs a material medium (air, water, steel)	Travels through vacuum
Vibration	Longitudinal (parallel to propagation)	Transverse (perpendicular to propagation)
Speed in air/vacuum	$\approx 340 \text{ m s}^{-1}$ in air	$c = 3.00 \times 10^8 \text{ m s}^{-1}$ in vacuum
Demonstration of transverse nature	None (cannot be polarised)	Polarisation

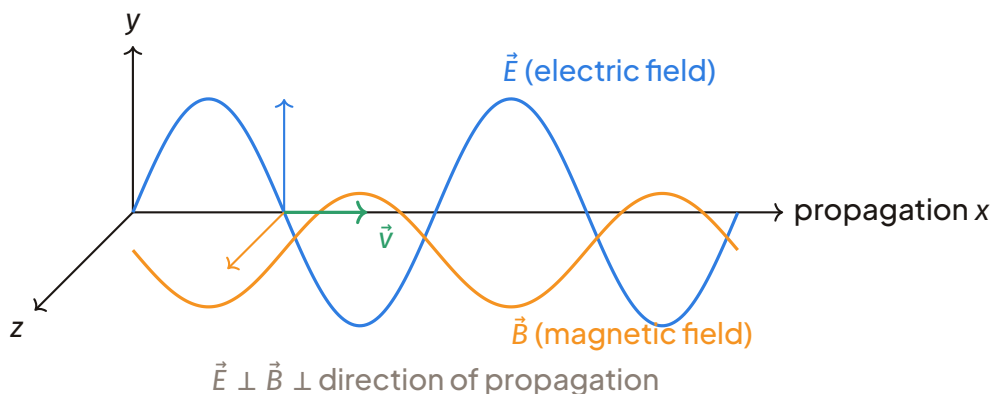
That light travels through the vacuum of space — we see the Sun and the stars — while sound does not (explosions in space films are pure fiction!) already tells us light is not a mechanical wave. What, then, is “waving”?

1.4 Maxwell’s Prediction and the Nature of Electromagnetic Waves

Around 1865, James Clerk Maxwell united the laws of electricity and magnetism into four equations and made a stunning prediction. A changing electric field creates a magnetic field (as a changing current does), and a changing magnetic field creates an electric field (electromagnetic induction). Maxwell realised these two processes can sustain each other: once an electric charge oscillates — say electrons surging up and down a transmitter aerial — it launches a self-propagating disturbance in which oscillating electric and magnetic fields continually regenerate one another, travelling outward even through empty space.

DEFINITION Electromagnetic wave

An **electromagnetic wave** is a self-propagating disturbance consisting of oscillating electric and magnetic fields, which requires no material medium. The electric field $\vec{E}$ and the magnetic field $\vec{B}$ vibrate perpendicular to each other and both perpendicular to the direction of propagation: the wave is **transverse**.



An electromagnetic wave: the electric field $\vec{E}$ and magnetic field $\vec{B}$ oscillate in phase, perpendicular to each other and to the direction of propagation.

Maxwell's equations also predicted the speed of these waves purely from electric and magnetic constants of vacuum:

$$c = \frac{1}{\sqrt{\epsilon_0 \mu_0}} \approx 3.00 \times 10^8 \text{ m s}^{-1},$$

exactly the measured speed of light. Maxwell concluded that **light itself is an electromagnetic wave** — one of the greatest unifications in the history of physics. Hertz confirmed the prediction experimentally about twenty years later by producing and detecting radio waves.

PROPERTY Speed of electromagnetic waves

All electromagnetic waves, from radio waves to gamma rays, travel in vacuum at the same speed

$$c = 3.00 \times 10^8 \text{ m s}^{-1},$$

and obey the universal wave relation

$$c = f\lambda.$$

Consequently, high-frequency waves have short wavelengths and low-frequency waves have long wavelengths.

EXAMPLE Wavelength of a CRTV radio signal

A radio station broadcasts at $f = 100 \text{ MHz} = 1.00 \times 10^8 \text{ Hz}$. Its wavelength in air (essentially vacuum) is

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{1.00 \times 10^8} = 3.00 \text{ m}.$$

This is why FM aerials are of the order of a metre long: the aerial must be comparable with the wavelength to absorb the wave efficiently.

1.5 The Transverse Nature of Light: a Preview of Polarisation

How do we know the oscillations of an EM wave are transverse? The decisive evidence is **polarisation**. A longitudinal wave, like sound, vibrates along the direction of travel: rotating a filter around the beam can never block it. But a transverse wave vibrates across the direction of travel, and a polarising filter (such as the lenses of polaroid sunglasses, or the polariser of a calculator display) transmits only the component of $\vec{E}$ aligned with its axis. Rotate a second filter through 90° and the light is completely extinguished — impossible for a longitudinal wave. Polarisation therefore proves that light is transverse, exactly as Maxwell's picture requires. We shall study polarisation quantitatively in Section 2.

KEY POINTS Key points of Section 1

- Every progressive wave obeys $v = f\lambda$.
- Echo method: $v = \frac{2nd}{t}$; the factor 2 comes from the return journey.
- Resonance tube: two successive resonances give $v = 2f(l_2 - l_1)$; subtracting eliminates the end correction.
- Sound is mechanical and longitudinal and needs a medium; light is electromagnetic and transverse and travels through vacuum.
- An EM wave consists of mutually perpendicular oscillating fields $\vec{E}$ and $\vec{B}$, both perpendicular to the direction of propagation.
- In vacuum all EM waves travel at $c = 3.00 \times 10^8 \text{ m s}^{-1}$ and obey $c = f\lambda$.
- Polarisation demonstrates the transverse nature of EM waves.

EXERCISE 1 Speed of sound by resonance

A resonance tube is used with a tuning fork of frequency 480 Hz. The first resonance occurs at $l_1 = 17.2 \text{ cm}$ and the second at $l_2 = 52.4 \text{ cm}$.

1. Calculate the wavelength of the sound in the tube.
2. Deduce the speed of sound in air.
3. Explain why using $\lambda = 4l_1$ would give a less accurate result.

SOLUTION Exercise 1

1. $\lambda = 2(l_2 - l_1) = 2(0.524 - 0.172) = 2 \times 0.352 = 0.704 \text{ m}$.

2. $v = f\lambda = 480 \times 0.704 \approx 338 \text{ m s}^{-1}$.
3. $\lambda = 4l_1$ ignores the end correction e : the antinode lies a small distance above the open end, so the true quarter-wavelength is $l_1 + e$, not l_1 . The result would be systematically too small. Using $(l_2 - l_1)$ cancels e exactly.

DID YOU KNOW? From Maxwell to mobile money

Every mobile phone call, every mobile-money transfer and every radio broadcast in Cameroon relies on Maxwell's 1865 prediction. The oscillating currents in a transmitter mast launch EM waves that regenerate themselves across hundreds of kilometres of empty space and air, until a tiny oscillating current is induced in your phone's aerial. The same equations that describe a CRTV signal also describe the light by which you read this page — only the frequency differs.

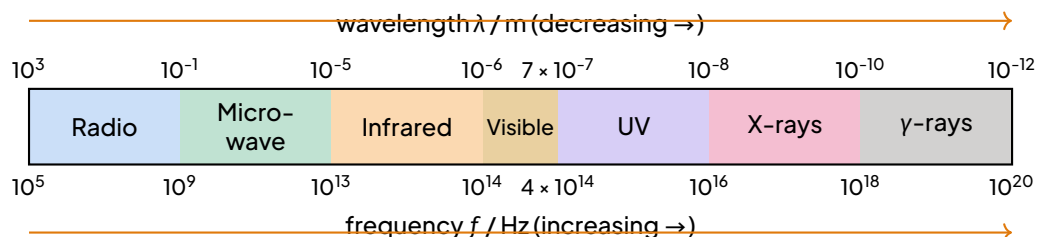
2 SECTION 2: THE ELECTROMAGNETIC SPECTRUM, X-RAYS AND POLARISATION

In Section 1 we established that light is only one member of a vast family of **electromagnetic waves** — self-sustaining oscillations of electric and magnetic fields that all travel through vacuum at the same speed $c = 3.00 \times 10^8 \text{ m s}^{-1}$ and obey the fundamental relation $c = f\lambda$. We also saw that sound, by contrast, is a mechanical longitudinal wave needing a material medium. In this section we exploit that difference fully: we organise the whole electromagnetic family into the **electromagnetic spectrum**, study how each region is produced, detected and used, examine in detail the production of X-rays in the Coolidge tube, and finish with a property that only transverse waves possess — **polarisation**.

2.1 The Electromagnetic Spectrum

Ordering the family. Since $c = f\lambda$ is fixed, the product $f\lambda$ is constant: a wave of long wavelength has a low frequency, and vice versa. Ordering all EM waves by decreasing wavelength (equivalently increasing frequency) gives the seven classical regions:

radio → microwave → infrared → visible → ultraviolet → X-rays → gamma rays



The electromagnetic spectrum. Wavelength decreases and frequency increases from radio waves to gamma rays; the visible region is only a narrow band (about 400–700 nm).

KEY POINTS Orders of magnitude to memorise

Region	Approximate wavelength	Approximate frequency
Radio waves	$> 0.1 \text{ m}$	$< 10^9 \text{ Hz}$
Microwaves	$10^{-3} \text{ to } 10^{-1} \text{ m}$	$10^9 \text{ to } 10^{11} \text{ Hz}$
Infrared	$7 \times 10^{-7} \text{ to } 10^{-3} \text{ m}$	$10^{11} \text{ to } 4 \times 10^{14} \text{ Hz}$
Visible light	$4 \times 10^{-7} \text{ to } 7 \times 10^{-7} \text{ m}$	$4 \times 10^{14} \text{ to } 7.5 \times 10^{14} \text{ Hz}$
Ultraviolet	$10^{-8} \text{ to } 4 \times 10^{-7} \text{ m}$	$7.5 \times 10^{14} \text{ to } 10^{16} \text{ Hz}$
X-rays	$10^{-12} \text{ to } 10^{-8} \text{ m}$	$10^{16} \text{ to } 10^{20} \text{ Hz}$
Gamma rays	$< 10^{-11} \text{ m}$	$> 10^{19} \text{ Hz}$

The boundaries overlap slightly; the regions are named according to how the waves are produced, not by sharp cut-offs.

Photon energy across the spectrum. Each EM wave can also be pictured as a stream of photons of energy

$$E = hf = \frac{hc}{\lambda},$$

where $h = 6.63 \times 10^{-34} \text{ J s}$ is Planck's constant. Since f increases from radio to gamma, photon energy increases across the spectrum in the same direction. Radio photons are far too weak to affect individual molecules; X-ray and gamma photons carry enough energy to ionise atoms — this single fact explains all the health hazards discussed below.

EXAMPLE Photon energy of visible and X-ray radiation

For green light, $\lambda = 500 \text{ nm}$:

$$E = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{5.00 \times 10^{-7}} = 3.98 \times 10^{-19} \text{ J} \approx 2.5 \text{ eV}.$$

For an X-ray of $\lambda = 0.10 \text{ nm}$: $E = 2.0 \times 10^{-15} \text{ J} \approx 12 \text{ keV}$ — about 5000 times more energetic.

Hence X-rays can eject electrons from atoms (ionisation) while visible light cannot.

2.2 Production, Detection and Uses of Each Region

Region	Production	Detection	Uses
Radio	Oscillating charges in transmitter aerials	Receiver aerial + tuned circuit	Radio and TV broadcasting (CRTV), radio astronomy
Microwave	Klystron / magnetron; electronic oscillators	Aerial, diode detector	Mobile phones, satellite links, radar, microwave ovens
Infrared	All warm bodies; heaters, lamps	Skin (warmth), thermistor, IR photodiode, thermal camera	Remote controls, thermal imaging, night vision, physiotherapy
Visible	Hot filaments, discharge tubes, LEDs, the Sun	The eye, photographic film, LDR, photodiode	Vision, photography, illumination, optical fibres
Ultraviolet	The Sun, mercury-vapour lamps, arcs	Fluorescent materials, photographic film, photodiode	Sterilisation, security marking, sun-tan lamps
X-rays	Sudden deceleration of fast electrons (Coolidge tube)	Photographic film, Geiger-Müller tube, solid-state detector	Medical radiography, airport security, crystal structure analysis
Gamma	Radioactive decay of unstable nuclei; nuclear reactions	Geiger-Müller tube, scintillation counter	Cancer radiotherapy, sterilisation of medical equipment, tracers

A few comments that examiners like to test:

- **Radio waves** are produced when electrons oscillate in a metal **aerial**; the same aerial, connected to a tuned circuit, detects them. CRTV broadcasts reach your receiver in Yaoundé exactly this way.
- **Microwave ovens** exploit resonance: microwaves of frequency about 2.45 GHz are strongly absorbed by water molecules, heating food from within.
- **Infrared** is emitted by every body above absolute zero; a thermal camera simply maps this emission, which is why rescue teams can locate people in the dark.
- **Ultraviolet** causes fluorescence: certain substances absorb UV and re-emit visible light. This is used to reveal security marks on banknotes (including the CFA franc) and to sterilise equipment, since UV kills bacteria.

METHOD Identifying an EM region from its use or detector (exam technique)

1. Read the clue: is it a use (oven, remote control, radiography), a detector (eye, GM tube) or a source (aerial, radioactive nucleus)?
2. Match the clue to the region using the table above.
3. If a wavelength or frequency is given, compute the partner quantity with $c = f\lambda$ and locate the region on the spectrum.
4. Always justify your answer in one sentence: “This is infrared because it is detected as heat by a thermistor.”

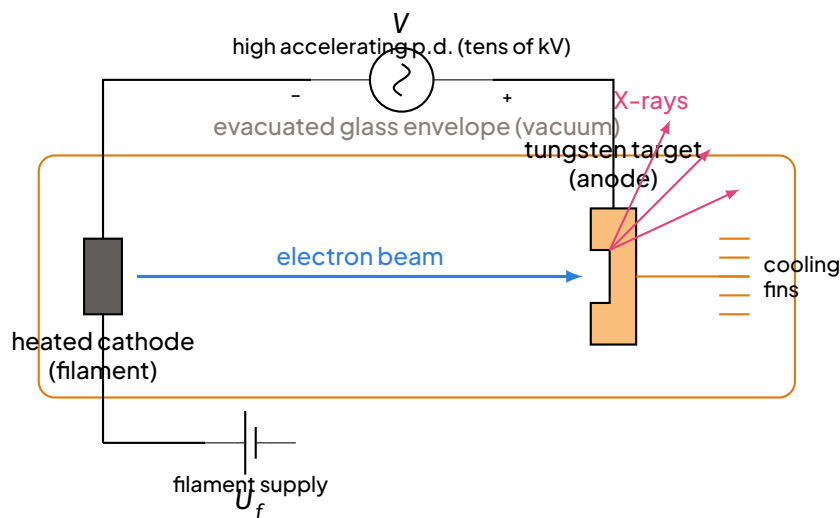
WARNING Ionising versus non-ionising radiation

Radio, microwave, infrared, visible and (mostly) ultraviolet photons are non-ionising: they can heat tissue or, for UV, damage skin (sunburn, skin cancer with over-exposure), but they cannot eject electrons from atoms. X-rays and gamma rays are ionising: they break chemical bonds inside living cells and can cause cancer. Hence radiographers stand behind lead screens, and radioactive sources are handled with tongs and stored in lead-lined containers. Never write that “all EM waves are dangerous” — precision earns marks.

2.3 X-rays: Production in the Coolidge Tube

PROPERTY Production of X-rays

X-rays are produced when fast-moving electrons are suddenly decelerated by striking a metal target. Part of the kinetic energy lost by the electrons is converted into electromagnetic radiation of very short wavelength (most of the rest heats the target). (The description of the tube is examinable; learn it with the diagram.)



The Coolidge (X-ray) tube. Electrons emitted by the heated cathode are accelerated through a large p.d. and decelerated by the tungsten target, emitting X-rays.

How the tube works, step by step.

1. A low-voltage supply heats the tungsten **filament** (cathode), which emits electrons by thermionic emission.
2. A very high p.d. (typically 20–100 kV) between cathode and anode accelerates the electrons to high speed. Each electron gains kinetic energy $E_k = eV$.
3. The electrons strike a **target** of a metal with a high melting point (tungsten) and are suddenly decelerated; a small fraction of their kinetic energy becomes X-ray photons (continuous spectrum / bremsstrahlung).
4. The tube is evacuated so that electrons are not scattered by air molecules.
5. Because about 99% of the energy appears as heat, the anode is fitted with cooling fins (or rotated and oil-cooled) to prevent melting.

The maximum photon energy corresponds to an electron losing all its kinetic energy in one collision (Duane–Hunt law):

$$eV = hf_{\max} = \frac{hc}{\lambda_{\min}} \implies \lambda_{\min} = \frac{hc}{eV}.$$

DID YOU KNOW? Characteristic X-rays

In addition to the continuous spectrum, if the accelerating voltage is high enough to eject inner-shell electrons from the target atoms, the subsequent de-excitation of outer electrons produces sharp characteristic X-ray lines (e.g. K_α , K_β) whose wavelengths depend only on the target material, not on the voltage. These are vital for X-ray fluorescence analysis.

WORKED EXAMPLE 2 Minimum wavelength from an X-ray tube

An X-ray tube operates at an accelerating p.d. of $V = 50 \text{ kV}$. Calculate (a) the kinetic energy of an electron reaching the target, (b) the minimum wavelength of the X-rays produced.

SOLUTION

(a) Kinetic energy gained by one electron:

$$E_k = eV = 1.60 \times 10^{-19} \times 5.0 \times 10^4 = 8.0e - 15 \text{ J.}$$

(b) The most energetic photon is produced when all of E_k becomes one photon:

$$\lambda_{\min} = \frac{hc}{eV} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{8.0 \times 10^{-15}} = 2.5e - 11 \text{ m.}$$

This lies in the X-ray region ($< 10^{-8} \text{ m}$), as expected.

Properties and uses of X-rays.

- They are EM waves of very short wavelength: they travel in straight lines, are not deflected by electric or magnetic fields (they carry no charge), and ionise gases.
- They penetrate matter, but are absorbed more by dense material: **bone absorbs X-rays far more than soft tissue**, which is why bones appear white on a radiograph.
- Uses: **medical radiography** (fractures, chest images), **airport security scanners** for luggage, and **X-ray diffraction by crystals**, which reveals the spacing of atomic planes.

WARNING X-rays $\neq$ gamma rays

A classic confusion. X-rays are produced outside the nucleus, by the sudden deceleration of fast electrons (or by electronic transitions in inner atomic shells). Gamma rays are emitted by unstable nuclei during radioactive decay, e.g. ${}^{60}_{27}\text{Co} \rightarrow {}^{60}_{28}\text{Ni} + {}^0_{-1}\text{e} + \bar{\nu}$. The two may have identical wavelengths — they differ by origin, not by nature. Saying “X-rays come from radioactive decay” loses the mark.

2.4 Plane Polarised Electromagnetic Waves

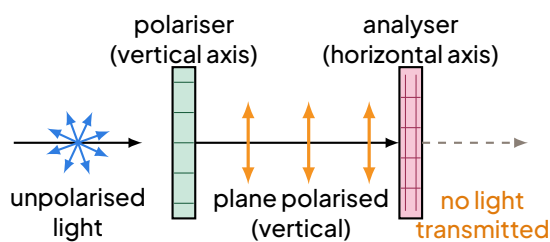
Intuition. A transverse wave on a rope can oscillate vertically, horizontally, or in any direction perpendicular to the rope. Ordinary light from a lamp or the Sun is a mixture of waves oscillating in all transverse directions at random: it is **unpolarised**. If we pass it through a filter that transmits only one direction of oscillation, the emerging wave oscillates in a single plane.

DEFINITION Plane polarised wave

A **plane polarised wave** is a transverse wave whose oscillations occur in one plane only — the plane containing the direction of propagation and the (single) direction of vibration. For an EM wave, it is the electric field vector $\vec{E}$ that is confined to this plane.

WARNING Only transverse waves can be polarised

Polarisation is the decisive experimental proof that light is a transverse wave. Sound in air is longitudinal: the oscillations are already along the direction of travel, so there is no transverse direction to select. Sound cannot be polarised. Writing “polarised sound” in the GCE is an automatic error.



Unpolarised light passes through a polaroid (the polariser); the emerging plane polarised light is completely blocked by a second polaroid (the analyser) whose transmission axis is crossed at 90° .

Malus-type behaviour (qualitative). Let plane polarised light of intensity I_0 fall on an analyser whose transmission axis makes an angle θ with the plane of polarisation. Only the component of $\vec{E}$ along the axis is transmitted, and since intensity is proportional to the square of the amplitude,

$$I = I_0 \cos^2 \theta.$$

Consequences to remember: $\theta = 0^\circ$ gives full transmission; $\theta = 90^\circ$ (crossed polaroids) gives darkness; rotating the analyser through 360° produces two maxima and two extinctions. Note also that when unpolarised light passes through a single polaroid, the transmitted intensity is exactly half, whatever the orientation.

EXAMPLE Intensity through two polaroids

Unpolarised light of intensity I_i falls on a polariser, then on an analyser at 60° to the polariser. After the polariser: $I_1 = \frac{1}{2} I_i$. After the analyser:

$$I_2 = I_1 \cos^2 60^\circ = \frac{1}{2} I_i \times (0.5)^2 = 0.125 I_i.$$

Only one eighth of the original intensity emerges.

Applications of polarisation.

- **Polaroid sunglasses:** light reflected by a road or by water is partially polarised horizontally; the vertical transmission axis of the lenses blocks this glare while transmitting useful light — a real comfort when driving on the Douala–Yaoundé highway in the sun.
- **Stress analysis:** a transparent plastic model placed between crossed polaroids shows coloured fringes where internal stresses are greatest; engineers use this to test bridges and machine parts before construction.
- **LCD screens:** a liquid-crystal display consists of a liquid-crystal layer between two crossed polaroids; applying a voltage rotates the plane of polarisation locally, switching each pixel between bright and dark.

DID YOU KNOW? Polarisation in everyday life

Bees navigate using the polarisation pattern of skylight, even when the Sun is hidden by clouds. Photographers, on the other hand, screw a polarising filter onto their lenses to darken blue skies and remove reflections from shop windows — the same physics as your sunglasses.

KEY POINTS GCE exam tip: one production, one detector, one use per region

Examiners regularly ask: “Name the region of the EM spectrum which ...” or “State one use and one detector of ...”. Memorise for each region one production mechanism, one detector and one use (the summary table above is enough). For X-rays, be able to draw and label the Coolidge tube and to state that X-rays arise from the sudden deceleration of fast electrons. For polarisation, remember the three keywords: transverse only, one plane, $I = I_0 \cos^2 \theta$.

EXERCISE 2 Regions, X-rays and polarisation

1. A wave has frequency $f = 6.0 \times 10^{14}$ Hz. Calculate its wavelength in vacuum and identify the region of the spectrum.
2. An X-ray tube operates at 30 kV. Calculate the maximum frequency of the X-rays produced.
3. Explain why the target of an X-ray tube is made of tungsten and why the tube must be evacuated.
4. Plane polarised light of intensity I_0 passes through an analyser. At what angle θ is the transmitted intensity equal to $\frac{1}{4}I_0$?

5. State, with a reason, whether sound waves in air can be polarised.

SOLUTION Exercise 1

1. $\lambda = c/f = (3.00 \times 10^8)/(6.0 \times 10^{14}) = 5.0 \times 10^{-7} \text{ m} = 500 \text{ nm}$: visible light (green).
2. $f_{\text{max}} = eV/h = (1.60 \times 10^{-19} \times 3.0 \times 10^4)/(6.63 \times 10^{-34}) = 7.2 \times 10^{18} \text{ Hz}$.
3. Tungsten has a very high melting point, so it resists the intense heating produced when 99% of the electron energy becomes heat. The vacuum prevents the electrons from colliding with air molecules, which would scatter them and reduce their energy before they reach the target.
4. $\cos^2 \theta = 1/4 \Rightarrow \cos \theta = 1/2 \Rightarrow \theta = 60^\circ$.
5. No. Sound in air is longitudinal: its oscillations are parallel to the direction of propagation, so there is no transverse plane to select. Only transverse waves can be polarised.

3 SECTION 3: LIGHT SOURCES, DIFFRACTION GRATINGS AND MULTIPLE-SLIT DIFFRACTION

In Section 2, we explored the vast electromagnetic spectrum and the production of X-rays. We also introduced the concept of polarisation, a property exclusive to transverse waves. Now, we turn our attention to the visible region of the spectrum — light. To understand how we analyse light, we must first understand where it comes from and how it behaves when it encounters a periodic structure with thousands of slits: the diffraction grating. This instrument is the workhorse of spectroscopy, allowing us to identify the chemical composition of stars, the gases in a neon sign, or the wavelength of a laser with remarkable precision.

3.1 Light Sources and the Nature of Spectra

Before studying diffraction, we must characterise the light sources we use in the laboratory and in nature.

DEFINITION Light Sources and Spectra

A **light source** is any object that emits electromagnetic radiation in the visible range ($\approx 400\text{--}700 \text{ nm}$). We classify them by their spectral output:

- Natural sources: The Sun (approximately a black body at 5800 K), stars, fireflies, lightning.

- Artificial sources: Incandescent filaments (tungsten lamps), gas discharge tubes (sodium, mercury, neon, hydrogen), Light Emitting Diodes (LEDs), Lasers.

The **spectrum** of a source is the distribution of intensity as a function of wavelength (or frequency).

- Continuous spectrum: Contains all wavelengths over a range with no gaps. Produced by hot, dense solids, liquids, or high-pressure gases (e.g., incandescent bulb filament, the Sun's photosphere).
- Line spectrum (Emission spectrum): Consists of bright, sharp lines at specific wavelengths separated by dark gaps. Produced by low-pressure gases in discharge tubes (e.g., sodium vapour lamp, neon sign). Each element has a unique "fingerprint" of lines.
- Absorption spectrum: A continuous spectrum crossed by dark lines at specific wavelengths. Produced when white light passes through a cooler, low-pressure gas (e.g., Fraunhofer lines in the solar spectrum).

DID YOU KNOW? Cameroon Context

In the physics laboratories of our lycées (e.g., Lycée Leclerc Yaoundé, GBHS Bamenda), the standard sources for diffraction experiments are the sodium vapour lamp (quasi-monochromatic yellow doublet at 589.0 and 589.6 nm) and the mercury vapour lamp (distinct lines: violet 435.8 nm, green 546.1 nm, yellow 577.0/579.1 nm). He-Ne lasers ($\lambda = 632.8$ nm) are increasingly common for demonstrating sharp diffraction orders.

3.2 Coherence: The Prerequisite for Interference and Diffraction

You know from double-slit interference that a stable pattern requires **coherent sources**. Two sources are coherent if they maintain a **constant phase difference** and have the **same frequency**.

KEY POINTS Conditions for Observable Interference/Diffraction Patterns

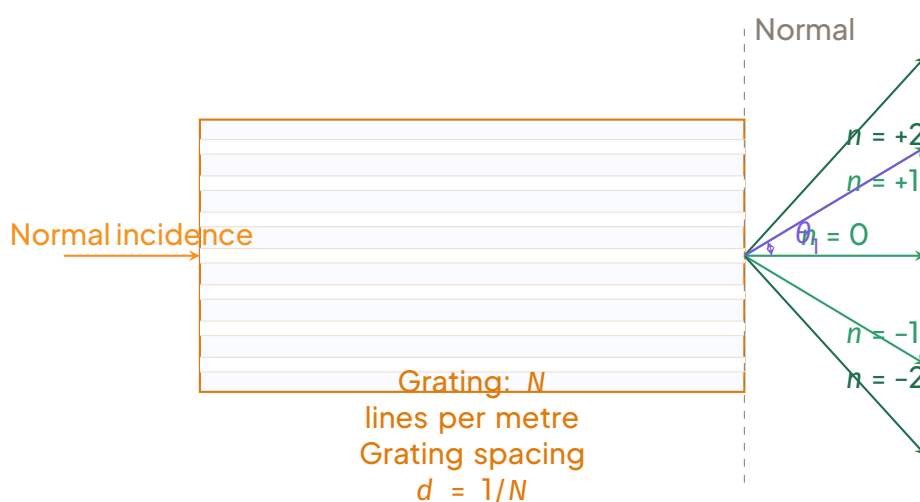
1. Monochromaticity (Same frequency): $\Delta f \approx 0$. White light produces overlapping patterns of different colours; only the central maximum ($n = 0$) remains white.
2. Constant phase relationship: The phase difference $\Delta\phi$ does not vary randomly with time.
3. Same polarisation state: For maximum contrast, the electric field vectors must oscillate in parallel planes.

In practice, we derive coherence from a single primary source (e.g., illuminating a grating

with light from one sodium lamp). The different slits of the grating then act as secondary coherent sources (Huygens' principle).

3.3 The Optical Transmission Grating

A transmission grating consists of a large number N of equally spaced, narrow, parallel slits (or transparent lines) ruled on a glass or plastic substrate.



Transmission grating at normal incidence. Parallel rays incident normally are diffracted into discrete orders $n = 0, \pm 1, \pm 2, \dots$ at angles θ_n satisfying $d \sin \theta_n = n\lambda$.

DEFINITION Grating Spacing

If the grating has N **lines per unit length** (usually given in lines per millimetre or lines per metre), the distance between the centres of two adjacent slits — the **grating spacing** or **grating element** — is:

$$d = \frac{1}{N}$$

SI Unit: metre (m). If N is in lines/mm, convert to lines/m first: $N_m = N_{\text{mm}} \times 10^3$.

WARNING Classic GCE Trap: Units of N and d

The syllabus and examiners frequently test unit conversion.

- A grating marked "500 lines/mm" means $N = 500 \times 10^3 = 5 \times 10^5$ lines/m.
- Then $d = \frac{1}{5 \times 10^5} = 2 \times 10^{-6} \text{ m} = 2000 \text{ nm}$.